

Hurricane Potential Intensity

Activity Sheet

Purpose

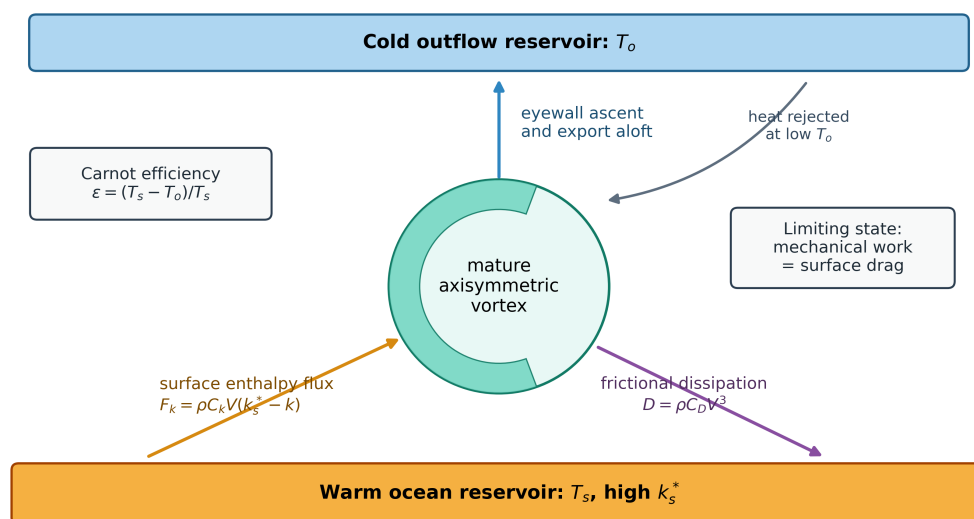
Use this sheet to organize your notes during the lecture and to answer conceptual questions about Kerry Emanuel's potential intensity theory.

1. Before the derivation: What should PI predict?

In one or two sentences, write what you think a “maximum possible hurricane intensity” theory should predict. Should it predict the intensity of every individual storm? Why or why not?

2. Heat-engine interpretation

The hurricane is idealized as a heat engine operating between the warm ocean surface and the cold upper-tropospheric outflow.



Question 2.1. Identify the warm reservoir, cold reservoir, heat input, and mechanical energy loss in the figure.

Warm reservoir

Cold reservoir

Heat input

Mechanical energy loss

Question 2.2. Why does a colder outflow temperature tend to increase PI?

3. Deriving the simple PI expression

The surface enthalpy flux is

$$F_k = \rho C_k V (k_s^* - k),$$

and the surface frictional dissipation rate is

$$D = \rho C_D V^3.$$

The Carnot efficiency without dissipative heating is

$$\epsilon = \frac{T_s - T_o}{T_s}.$$

Question 3.1. At limiting steady intensity, assume

$$D = \epsilon F_k.$$

Use this to derive

$$V^2 = \frac{C_k}{C_D} \frac{T_s - T_o}{T_s} (k_s^* - k).$$

Question 3.2. With dissipative heating, the common modern form is

$$V_p^2 = \frac{C_k}{C_D} \frac{T_s - T_o}{T_o} (k_s^* - k).$$

Which denominator changed, and why does that increase the predicted intensity?

4. Physical interpretation of each term

For each term, state whether increasing the term should increase, decrease, or have ambiguous effect on PI. Then explain the physical reason.

Term	Effect on V_p	Physical explanation
C_k/C_D		
$(T_s - T_o)/T_o$		
$k_s^* - k$		

5. Numerical calculation

Assume

$$T_s = 300 \text{ K}, \quad T_o = 200 \text{ K}, \quad k_s^* - k = 20,000 \text{ J kg}^{-1},$$

and

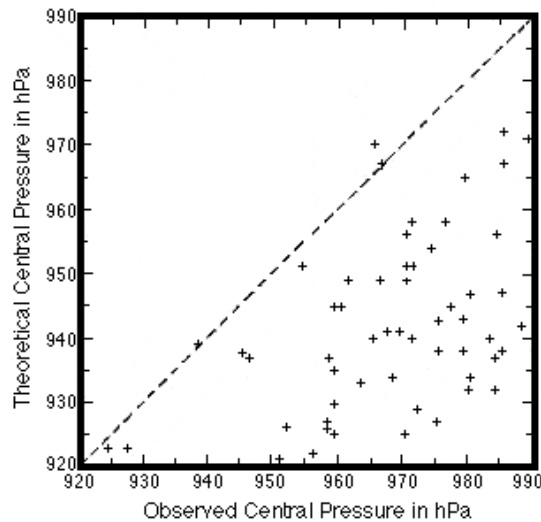
$$\frac{C_k}{C_D} = 1.$$

Question 5.1. Calculate V_p using

$$V_p^2 = \frac{C_k}{C_D} \frac{T_s - T_o}{T_o} (k_s^* - k).$$

Question 5.2. Suppose vertical shear is strong. Does the calculated V_p change in the formula above? Does the actual storm intensity likely change? Explain the distinction.

6. Observational testing: upper bound or prediction?



. Source of Image:
<https://emanuel.mit.edu/hurricane-potential-intensity-maps-0/>

Question 6.1. In an observational plot of V_{max} versus V_p , what would it mean if most points lie below the one-to-one line?

Question 6.2. Why should we not expect every observed storm to lie close to the line $V_{\text{max}} = V_p$?

7. Critique: why actual intensity differs from PI

Recall three limitations of the potential intensity theory. For each one, describe how it could prevent a storm from reaching PI.

Limitation	How it affects actual intensity
1.	
2.	
3.	

8. Synthesis question

A storm is located over warm SST and has a high calculated PI. However, it is moving slowly over a shallow mixed layer and is experiencing moderate vertical wind shear.

Question. Would you expect this storm to reach its PI? Use at least two physical arguments.

9. Minute Reflection

In one sentence, state the most defensible interpretation of Emanuel PI.
